Khanh Nguyen, Postdoc, Ph.D.

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EDUCATION

Doctor of Philosophy in Smart Vehicle Engineering
Konkuk University, Seoul, Republic of Korea | 2022 – 2026 | **GPA: 4.0/4.0**
Dissertation: *Scaling-based design approach for tailbeat fish-inspired robots swimming at high speed and propulsive efficiency*

Master of Science in Smart Vehicle Engineering
Konkuk University, Seoul, Republic of Korea | 2019 – 2021 | **GPA: 3.9/4.0**
Thesis: *Investigation of stability and aerodynamic performance of a flapping-wing micro air vehicle in hover using 3D computational fluid dynamics analyses*

Bachelor of Engineering in Mechanical and Aerospace Engineering
Vietnam National University, Ho Chi Minh City University of Technology, Vietnam | 2013 – 2018 | **GPA: 3.2/4.0**
The five-year France–Vietnam Training Program of Excellent Engineers (PFIEV) is accredited by France's Engineering Degree Commission and designated as a Master Program by the European Network for Accreditation of Engineering Education.
Thesis: *Computational approach on the aerodynamics of UAV combining fixed wing and three propellers*
Grade: 9.1/10

RESEARCH EXPERIENCE

Postdoctoral Researcher – Konkuk University (03/2026 – Present)
Analyze the forward flight kinematics of a tethered flapping-wing robot under various inflow velocities.
Validate the proposed scaling-based design approach by developing a larger fish robot capable of swimming at 2.5 m/s.
Analyze the effect of four-bar linkage on the tailbeat frequency of the robotic tailbeat mechanism.
Investigate vortex generation and force production by the caudal fin using measured kinematics from free-swimming robot.

Research Co-Supervisor (Remote, Non-tenure-track) – Ho Chi Minh University of Technology (01/2026 – Present)
Develop research directions and guide numerical simulations using user-defined-functions in ANSYS-Fluent.
Co-supervise research in bio-inspired hydrodynamics and aerodynamics, focusing on efficiency and propulsion.

Research Assistant – Konkuk University (03/2019 – 02/2026)
Developed a fish robot capable of efficiently swimming at 1.79 m/s using a scaling-based design approach. (1 SCIE paper)
Investigated characteristics of dynamic flight stability and aerodynamics of a flapping-wing hovering on Mars. (2 SCIE papers)
Analyzed the feasibility of the gliding flight in a flying-fish-like robot after water exit. (1 SCIE paper)
Estimated body drag of undulatory robotic fish using the measured swimming kinematics. (1 SCIE paper)
Investigated the aerodynamic effects of spanwise corrugation, wing rotation angles, and clap-fling mechanism. (1 SCIE paper)
Compared the longitudinal and lateral stability characteristics of two flapping-wing robots. (1 SCIE paper)

Research Assistant – Ho Chi Minh City University of Technology (02/2018 – 02/2019)
Estimated the tip-path-plane angle of the forward-flight tricopter using CFD-derived force coefficients (1 SCOPUS paper).
Analyzed aerodynamics of a tri-copter frame during takeoff using the Virtual Blade Method in OpenFOAM.

MENTORSHIP EXPERIENCE

- Quoc Anh Le (Undergraduate student, Ho Chi Minh city University of Technology, Sept. 2026 – Present)
  Project: Effects of free-surface proximity and chordwise morphing on the hydrodynamic efficiency of a flapping foil
- Tan Nhat Bui (Undergraduate student, Ho Chi Minh city University of Technology, Sept. 2026 – Present)
  Project: Aerodynamic studies of small robotic flyer powered by a cyclogyro propulsion system
- An Thien Le (Undergraduate student, Ho Chi Minh city University of Technology, Sept. 2026 – Present)
  Project: Eddy blocking mechanism for noise and drag reduction via high-frequency trailing-edge vibration
- Dan Thanh Tran (Master student, Pukyong National University, Aug. 2026 – Present)
  Project: Role of Slats, Multi-Slot Configurations, and Tesla Valve-Inspired Designs in Improving Flapping-Foil Efficiency
- Minh Khoa Le (Undergraduate student, Ho Chi Minh city University of Technology, Jan. 2026 – Present)
  Project: Optimization of flapping and chordwise morphing hydrofoil for enhanced hydrodynamic force control
- Lim Sang-Hyeon (Master student, Konkuk University, Dec. 2025 – Present)

Project: Modeling and control design for an underwater robotic fish
- Doan Tran Kim Khanh (Undergraduate student, Ho Chi Minh city University of Technology, Jan. 2019 – Jun. 2019)
Project: OpenFOAM simulation of a tricopter in forward flight using the virtual blade model.
Outcome: 01 Scopus-indexed paper

SERVICES

Professional Services

Journal Reviewer

- *Bioinspired Robotics & Marine Engineering:* Ocean Engineering (Q1), Bioinspiration & Biomimetics (Q1).
- *Aerodynamics & Fluid Mechanics:* Chinese Journal of Aeronautics (Q1), Journal of Fluids and Structures (Q1), Journal of Aeronautics, Astronautics and Aviation (Q3).
- *Computational Engineering & Intelligent Systems:* Journal of Computational Design and Engineering (Q1), Engineering Research Express (Q2), International Journal of Intelligent Unmanned Systems (Q4).

Academic Services & Teaching Assistant

Grading Assistant, Basics of Mechanics & Finite Element Method (KU); Fluid Mechanics (HCMUT).
Thesis Evaluator, Bachelor's graduation theses (HCMUT).

AWARDS & HONORS

Outstanding Dissertation, Korean Society of Mechanical Engineers (Sole foreign awardee among nationwide Top 5; 2026)
Outstanding Research Achievement, Academic Award for the best Ph.D. Graduate, College of Engineering, KU (2026)
Best Paper, International Conference on Intelligent Unmanned Systems (2022, 2025)
Best Paper, Korea Society for Aeronautical and Space Sciences Conference (2024)
Excellent Student, Ho Chi Minh City University of Technology (150% Tuition Award; 2018)
Quintessential Student, Ho Chi Minh City University of Technology (125% Tuition Award; 2017, 2014)
Quintessential Student, Tran Phu High School (National University Entrance Exam; Score: 25/30, Top 1% Nationwide; 2013)

FELLOWSHIPS & ACADEMIC APPOINTMENTS

Brain Korea 21 Postdoctoral Scholarship, Korean Government ($23,000; 2026)
Postdoctoral Fellowship, Konkuk University ($33,000; 2026)
Ph.D. Fellowship, Konkuk University ($78,000; 2022 – 2026)
Research Assistant, Ho Chi Minh City University of Technology (2018); Konkuk University (2019 – 2021). Total: $23,200
Teaching Assistant, Ho Chi Minh City University of Technology (2018); Konkuk University (2024). Total: $10,100

RESEARCH PROJECTS

Unsteady fluid dynamics of oscillating foils (Jan. 2026 – Current)

Funding source: Partially Brain Korea 21 Program and Defense Acquisition Program Administration
Principal Investigator: Khanh Nguyen
My Role: Key Personnel – CFD Simulation & Bioinspired Propulsion Development
Outcome: N/A (As of Oct 2026)
This is my independent, self-initiated research project, which aims to investigate the mechanisms underlying animal locomotion and apply these principles observed in nature to bio-inspired surface vehicles powered by actively flexible flapping foils. When operated along a controllable inclined stroke plane, a submerged flapping-foil system can generate both thrust and vertical force, enabling the vehicle to move forward while supporting vehicles with varying weights.

Development of a biomimetic underwater robot platform (Mar. 2023 – Feb. 2028)

Funding source: Defense Acquisition Program Administration
Principal Investigator: Jihoon Kim
My Role: Key Personnel – Robotic Design, Prototyping & Performance Optimization
Outcome: 01 SCIE paper (As of Oct 2026)
This research proposes a scaling-based approach for designing robotic fish to achieve predefined maximum swimming speeds. I was responsible for mechanical design, fabrication, experimental testing, and numerical simulations to optimize body and tail undulation. Using this approach, I developed the 50-cm KUFish-III, achieving 2.5 m/s, and the 32-cm KUFish-II, achieving 1.8 m/s (Froude number = 1.0) with a Strouhal number of 0.28 and a tailbeat amplitude of 13% of body length. These results demonstrate high-speed, energy-efficient swimming with a low cost of transport.

Mars Atmospheric Flying Robot Laboratory (Jun. 2022 – Feb. 2025)

Funding source: National Research Foundation of Korea

Principal Investigator: Hoon Cheol Park

My Role: Key Personnel – CFD Simulation & Flight Stability Analysis

Outcome: 02 SCIE papers (As of Oct 2026)

This research aims to develop a flapping-wing robot capable of taking off under simulated Martian conditions using a pressure-controlled chamber. I investigated flight stability and hovering responses to small disturbances under Martian and terrestrial conditions. I also conducted computational fluid dynamics (CFD) simulations using measured wing kinematics to evaluate aerodynamic characteristics under varying low-density conditions.

Development of key technologies for a flying-fish-inspired robot (Jun. 2020 – May 2023)

Funding source: National Research Foundation of Korea

Principal Investigator: Hoon Cheol Park

My Role: Hydrodynamic & Aerodynamic Simulation and Performance Analysis

Outcome: 02 SCIE papers

This research investigated the feasibility of a fish-inspired robot transitioning from high-speed underwater swimming to aerial gliding under a zero-pitching condition. I conducted numerical simulations to evaluate the hydrodynamic characteristics using measured swimming kinematics and assess gliding performance under headwind conditions. I identified suitable body, wing, and tail tilt angles that generate sufficient thrust to overcome drag and lift to balance the robot's weight. These findings provide a basis for developing fish-inspired robots capable of underwater and aerial locomotion.

Vortex Physics-driven Flight Innovation (Dec. 2018 – Dec. 2020)

Funding source: National Research Foundation of Korea

Principal Investigator: Hoon Cheol Park

My Role: CFD Simulation & Aerodynamic Performance Optimization

Outcome: 01 SCIE paper

In this research, I used computational fluid dynamics (CFD) simulations to investigate the effects of wing kinematics, corrugation structures, and clap-and-fling motion on the aerodynamic efficiency of a hovering flapping-wing micro air vehicle (KUBeetle). I evaluated modified wing kinematics incorporating spanwise twist and chordwise camber, as well as different wing corrugation structures and clap-and-fling motion. The results showed that optimized wing kinematics with distributed corrugations, without clap-and-fling motion, improved the lift-to-drag ratio while maintaining sufficient lift. These findings provide design guidelines for improving the aerodynamic efficiency and flight endurance of flapping-wing micro air vehicles.

Basic study for flapping-wing micro air vehicles (Dec. 2013 – Dec. 2021)

Funding source: Defense Acquisition Program Administration and the Agency for Defense Development

Principal Investigator: Hoon Cheol Park

My Role: CFD Simulation & Flight Dynamics Analysis

Outcome: 01 SCIE paper

In this research, I compared the hovering stability of two insect-inspired flapping-wing micro air vehicles (KUBeetle-SPC and KUBeetle-TEC) employing different attitude control mechanisms. I used computational fluid dynamics (CFD) simulations to estimate aerodynamic forces and moments and analyzed their longitudinal and lateral dynamic stability. The results showed that the SPC exhibited faster stabilization of subsidence modes but greater instability in divergence modes than the TEC. I also investigated the effects of aerodynamic stability derivatives to identify the primary sources of instability. The SPC demonstrated faster responses to external disturbances, suggesting improved maneuverability in cluttered environments.

TECHNICAL SKILLS

Languages: C, Python, HTML/CSS.

Development Tools: MATLAB, Visual Studio, VS Code.

Simulation and Post-Processing Tools: ANSYS-Fluent, CFD-Post, OpenFOAM, ParaFoam.

Tools: CNC & 3D Printing Machines, Direct Linear Transformation Digitizing Tool.

Software: AutoCAD, Adobe Photoshop, SolidWorks.

Processes: Silicone Mold Making, Transducer Measurement, Image Processing.

PUBLICATIONS

Journal Papers (Rankings based on Clarivate's Journal Citation Reports – JCR; *Corresponding author)

1. Khanh Nguyen, Hoon Cheol Park*. Scaling-based design approach for tailbeat fish-inspired robots swimming at high speed and propulsive efficiency. Ocean Engineering, 368, 128377, 2026. (Q1, IF = 6.3, Rank: 1/26 in Marine Engineering)

2. Giheon Ha, Khanh Nguyen, Jiyeon Yu, Teasam Kang, Hoon Cheol Park*. Aerodynamic characteristics of flapping wings at takeoff under very low air density and low gravity. Acta Astronautica, 2026. (Q1, IF = 3.7, Top 17% in Aeros. Engineering)
3. Thi Hong Hieu Le*, Khanh Nguyen, Vuong Thi Hong Nhi. Numerical analysis for aerodynamic characteristics of the unmanned aerial vehicle (UAV) in forward flight. J. Aeronaut. Astronaut. Aviat. 2024. (Q3, IF = 1.1)
4. Khanh Nguyen, Giheon Ha, Teasam Kang, Hoon Cheol Park*. Analysis of hovering flight stability of an insect-like flapping-wing robot in Martian condition. Aerosp. Sci. Technol. 2024. (Q1, IF = 5.8, Top 10% in Aerospace Engineering)
5. Khanh Nguyen, Hoon Cheol Park*, Feasibility study on mimicking the tail-beating supported gliding flight of flying fish. Ocean Engineering 2023. (Q1, IF = 4.6, Rank: 2/25, Top 8% in Marine Engineering)
6. Tan Hanh Pham, Khanh Nguyen, Hoon Cheol Park*. A robotic fish capable of fast underwater swimming and water leaping with high Froude number. Ocean Engineering 2023. (Q1, IF = 4.6, Rank: 2/25, Top 8% in Marine Engineering)
7. Khanh Nguyen, Loan Thi Kim Au, Hoang Vu Phan, Hoon Cheol Park*. Comparative dynamic flight stability of insect-inspired flapping-wing micro air vehicles in hover: Longitudinal and lateral motions. Aerosp. Sci. Technol. 2021. (Q1, IF = 5.5, Rank: 2/34, Top 6% in Aerospace Engineering)
8. Khanh Nguyen, Loan Thi Kim Au, Hoang-Vu Phan, Soo Hyung Park, Hoon Cheol Park*. Effects of wing kinematics, corrugation, and clap-and-fling on aerodynamic efficiency of a hovering insect-inspired flapping-wing micro air vehicle. Aerosp. Sci. Technol. 2021. (Q1, IF = 5.5, Rank: 2/34, Top 6% in Aerospace Engineering)
9. Doan Kim Khanh Tran, Khanh Nguyen, Thi Hong Hieu Le, Ngoc Hien Nguyen*. Numerical simulation for the forward flight of the tri-copter using virtual blade model. J. Adv. Res. Fluid Mech. Therm. Sci. 2020. (Scopus Q3)

Conference Papers (*Presenter, **Corresponding author)

1. Khanh Nguyen*, Giheon Ha, Hoon Cheol Park**, Design and demonstration of a high-speed aquatic swimmer using tail-beat propulsion, International Conference on Intelligent Unmanned Systems, Indonesia, 2025 (Best paper).
2. Khanh Nguyen*, Giheon Ha, Hoon Cheol Park**, Design and fabrication of high-thrust tail-beating mechanism for fish-inspired swimming robot, International Conference on Intelligent Unmanned Systems, Indonesia, 2024.
3. Khanh Nguyen*, Hoon Cheol Park**, Analytical and experimental performance verifications of a fast-swimming robotic fish, International Conference on Intelligent Unmanned Systems, Indonesia, 2024.
4. Khanh Nguyen*, Kang Teasam, Hoon Cheol Park**, Hovering characteristics of an insect-like flapping-wing robot on Mars, Korean Society for Aeronautical & Space Sciences, 2023.
5. Khanh Nguyen, Giheon Ha, Hoon Cheol Park*,**, Preliminary design of a fish-like fast robot by scaling of the KUFish, International Conference on Intelligent Unmanned Systems, Adelaide, Australia, 2023.
6. Khanh Nguyen*, Hoon Cheol Park, Roles of hydrodynamic forces generated by tail-beating motion in gliding flight of flying-fish-mimicking robot, International Conference on Intelligent Unmanned Systems, Adelaide, Australia, 2023.
7. Khanh Nguyen*, Tan Hanh Pham, Hoon Cheol Park**, Numerical investigation of hydrodynamics for a fish-like robot under undulatory forward swimming, Korean Society of Mechanical Engineers Annual Meeting, 2022.
8. Tan Hanh Pham*, Khanh Nguyen, Hoon Cheol Park**, Leaping out of water of the KUFish: Prediction and demonstration, International Conference on Intelligent Unmanned Systems, Japan, 2022. (Best paper).
9. Khanh Nguyen*, Tan Hanh Pham, Hoon Cheol Park**, Numerical estimation of hydrodynamic thrust using the measured tail-beating kinematics of a fish-like robot, International Conference on Intelligent Unmanned Systems, Japan, 2022.
10. Khanh Nguyen*, Loan Thi Kim Au, Hoang Vu Phan, Hoon Cheol Park**, Wing kinematics modulation in an insect-like tailless flapping wing micro air vehicle for higher aerodynamic efficiency, International Conference on Intelligent Unmanned Systems, Vietnam, 2021.
11. Khanh Nguyen*, Loan Thi Kim Au, Hoon Cheol Park**, Three-dimensional wing kinematics for improved aerodynamic performance of insect-like flapping-wing micro air vehicle, Korean Society for Aeronautical & Space Sciences, 2020.
12. Loan Thi Kim Au, Khanh Nguyen*, Soo Huyn Park, Hoon Cheol Park**, Effect of wing corrugation on aerodynamic performance in 3D flapping wings, Korean Society for Aeronautical & Space Sciences, 2019.
13. Doan Kim Khanh Tran*, Khanh Nguyen, Thi Hong Hieu Le, Numerical simulation for the forward flight of the tri-copter using Virtual Blade Model, Southeast Asia Workshop on Aerospace Engineering, Malaysia, 2019.
14. Khanh Nguyen*, Ngoc Hien Nguyen, Thi Hong Hieu Le**, Numerical approach for the vertical take-off and landing UAVs using the virtual blade model, Southeast Asia Workshop on Aerospace Engineering, Thailand, 2018.